WoSM Academy

MAT231-F27 · CALCULUS II

Constrained optimisation

Lagrange multipliers

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Last time

- Gradient descent: $\mathbf{x}_{k+1} = \mathbf{x}_k - \eta \nabla f(\mathbf{x}_k)$.
- On a convex quadratic it converges iff $0 < \eta < 2/L$; with the best fixed step the error shrinks by $(\kappa - 1)/(\kappa + 1)$ per step.
- Ill-conditioned problems zigzag; in general we only reach a critical point.

So far the variables were free. Today they must satisfy a constraint — a budget, a fixed total, a surface to stay on — and the condition $\nabla f = \mathbf{0}$ is replaced by $\nabla f = \lambda \nabla g$.

This lecture

01 Constrained problems

02 The method of Lagrange multipliers

03 What the multiplier means

04 Applications

05 More constraints, and limits of the method

06 Summary and practice

Reading: Strang & Herman, *Calculus, Volume 3*, Section 4.8; Marsden & Tromba, *Vector Calculus*, Section 3.4.

01

Constrained problems

Optimisation with a constraint

The problem

Find the largest or smallest value of $f(\mathbf{x})$ **subject to** $g(\mathbf{x}) = c$, that is, on the level set $S = \{\mathbf{x} : g(\mathbf{x}) = c\}$.

- The boundary step of Lecture 14: the extrema of f on a circle or a sphere.
- Geometry: the point of a surface closest to a given point.
- Economics: the largest output, or utility, for a fixed budget.
- Information theory: the distribution of largest entropy, among $p_i \geq 0$ with $\sum_i p_i = 1$.

Critical points of f on $\mathbb{R}^n$ do not help: the extrema on S are usually not critical points of f.

First method: parametrise the constraint

Example ($f(x,y) = x + y$ on the circle $x^2 + y^2 = 1$)

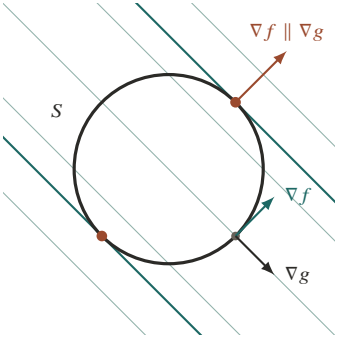
Put $x = \cos t$, $y = \sin t$:

$$f = \cos t + \sin t = \sqrt{2}\,\sin\!\left(t + \frac{\pi}{4}\right).$$

Maximum $\sqrt{2}$ at $t = \frac{\pi}{4}$, the point $\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$; minimum $-\sqrt{2}$ at the opposite point.

- This turns the problem into one of fewer variables, without constraints.
- But most constraint sets have no convenient parametrisation: try $x^4 + y^4 + z^4 = 1$, or $\sum_i p_i = 1$ with a thousand p_i.

The geometric picture



Level lines $x + y = k$ and the circle S.

Walk along S and watch the level lines of f you cross.

- Where a level line **crosses** S, f still increases one way along S.
- At an extremum the level line just **touches** S: the two curves are tangent.
- Tangent curves have parallel normals, and ∇f, ∇g are the normals (Lecture 11).

$$\nabla f = \lambda \nabla g.$$

02

The method of Lagrange multipliers

Lagrange's theorem

Theorem (Lagrange)

Let f, g be C^1 on an open set $U \subseteq \mathbb{R}^n$, and $S = \{\mathbf{x} \in U : g(\mathbf{x}) = c\}$. If f restricted to S has a local extremum at $\mathbf{a} \in S$ and $\nabla g(\mathbf{a}) \neq \mathbf{0}$, then

$$\nabla f(\mathbf{a}) = \lambda \nabla g(\mathbf{a}) \qquad \text{for some } \lambda \in \mathbb{R}.$$

- λ is the **Lagrange multiplier**.
- With the **Lagrangian** $\mathcal{L}(\mathbf{x}, \lambda) = f(\mathbf{x}) - \lambda\,\big(g(\mathbf{x}) - c\big)$, the conditions say that $(\mathbf{a}, \lambda)$ is a critical point of $\mathcal{L}$: $\nabla_{\mathbf{x}} \mathcal{L} = \nabla f - \lambda \nabla g = \mathbf{0}$ and $\partial \mathcal{L}/\partial \lambda = c - g = 0$.
- $n + 1$ equations for the $n + 1$ unknowns $x_1, \ldots, x_n, \lambda$.

Why it holds

Let $\gamma(t)$ be a C^1 curve in S with $\gamma(0) = \mathbf{a}$, and $\mathbf{v} = \gamma'(0)$.

- $g(\gamma(t)) = c$ for all t, so by the chain rule (Lecture 12) $\nabla g(\mathbf{a}) \cdot \mathbf{v} = 0$.
- $f(\gamma(t))$ has a local extremum at $t = 0$, so $\nabla f(\mathbf{a}) \cdot \mathbf{v} = 0$.

So $\nabla f(\mathbf{a})$ is perpendicular to every velocity $\mathbf{v}$ of a curve in S through $\mathbf{a}$.

> **The one fact we borrow**
>
> If $\nabla g(\mathbf{a}) \neq \mathbf{0}$, then **every** $\mathbf{v}$ with $\nabla g(\mathbf{a}) \cdot \mathbf{v} = 0$ is the velocity of such a curve: near $\mathbf{a}$, S is a smooth hypersurface with tangent space $\nabla g(\mathbf{a})^{\perp}$. This follows from the implicit function theorem (Lecture 17).

Then $\nabla f(\mathbf{a}) \perp \nabla g(\mathbf{a})^{\perp}$, so $\nabla f(\mathbf{a})$ lies in the line spanned by $\nabla g(\mathbf{a})$. □

The recipe

To find the extrema of f on $S = \{g = c\}$

1. Solve $\nabla f = \lambda \nabla g,\ \ g = c$ for $\mathbf{x}$ and λ.
2. Add the points of S where $\nabla g = \mathbf{0}$.
3. If S is closed and bounded, f has a maximum and a minimum on S (Lecture 9): evaluate f at all candidates and compare.

- The equations are usually nonlinear. Useful moves: divide two equations to eliminate λ; split into cases ($x = 0$ or $x \neq 0$); use symmetry.
- The value of λ is often not needed — but it has a meaning (later today).
- If S is not bounded, a separate argument is needed that an extremum exists.

Example: $x + y$ on the unit circle, again

$f = x + y$, $g = x^2 + y^2 = 1$:

$$\nabla f = (1, 1) = \lambda\,(2x, 2y) = \lambda \nabla g.$$

- $\lambda \neq 0$ (else $1 = 0$), so $x = y = \frac{1}{2\lambda}$.
- Then $x^2 + y^2 = 2x^2 = 1$: $x = y = \pm\frac{1}{\sqrt{2}}$, with $\lambda = \pm\frac{1}{\sqrt{2}}$.
- $\nabla g = (2x, 2y) \neq \mathbf{0}$ on the circle.

The circle is closed and bounded, so compare: $f\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right) = \sqrt{2}$ is the maximum, $f\left(-\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right) = -\sqrt{2}$ the minimum. As before.

Example: the best rectangle

Example (Largest area for a given perimeter P)

Maximise $A = xy$ subject to $2x + 2y = P$, with $x, y \geq 0$.

$$(y, x) = \lambda\,(2, 2) \quad \Longrightarrow \quad x = y = 2\lambda.$$

The constraint gives $x = y = \frac{P}{4}$ and $\lambda = \frac{P}{8}$. Area $\frac{P^2}{16}$.
The constraint set is a segment; at its endpoints ($x = 0$ or $y = 0$) the area is 0. So the maximum is the **square**.

Example: the closest points of a hyperbola

Example (Closest points of $xy = 1$ to the origin)

Minimise $x^2 + y^2$ subject to $xy = 1$: $(2x, 2y) = \lambda\,(y, x)$.

- Multiplying the two equations: $4xy = \lambda^2 xy$, and $xy = 1$, so $\lambda = \pm 2$.
- $\lambda = 2$: $x = y$, so $x^2 = 1$: the points $(1, 1)$ and $(-1, -1)$.
- $\lambda = -2$: $x = -y$, and $xy = -x^2 = 1$ is impossible.

The closest points are $\pm(1, 1)$, at distance $\sqrt{2}$.

The hyperbola is not bounded, so existence needs an argument: $x^2 + y^2 \to \infty$ along both ends of each branch, so a minimum exists, and it must be among the candidates.

Example: a product with a fixed sum

Example (Maximise xyz subject to $x + y + z = 3$, $x, y, z \geq 0$)

$(yz,\ xz,\ xy) = \lambda\,(1, 1, 1)$. If all three are positive: $yz = xz$ gives $x = y$, and $xz = xy$ gives $y = z$. So $x = y = z = 1$, with product 1.

If one of them is 0, the product is 0. The set is closed and bounded, so the maximum is 1.

The AM–GM inequality

Rescaling to $x + y + z = 3$: for all $x, y, z \geq 0$, $\sqrt[3]{xyz} \leq \dfrac{x + y + z}{3}$. The same method gives the inequality for n numbers.

03

What the multiplier means

λ is the rate of change of the optimum

Let $M(c)$ be the optimal value of f on $\{g = c\}$, attained at $\mathbf{a}(c)$ with multiplier $\lambda(c)$. Assume $\mathbf{a}(c)$ is differentiable in c.

> **Theorem**
> $M'(c) = \lambda(c)$.

Proof. $M(c) = f(\mathbf{a}(c))$ and $g(\mathbf{a}(c)) = c$, so by the chain rule

$$M'(c) = \nabla f(\mathbf{a}) \cdot \mathbf{a}'(c) = \lambda \, \nabla g(\mathbf{a}) \cdot \mathbf{a}'(c) = \lambda \, \frac{d}{dc} \, g(\mathbf{a}(c)) = \lambda. \qquad \square$$

Rectangle: $M(P) = \frac{P^2}{16}$ and $M'(P) = \frac{P}{8} = \lambda$: one more unit of fence buys about $\frac{P}{8}$ more area. Economists call λ a **shadow price**: the value, at the optimum, of one more unit of the constrained resource.

04

Applications

Maximum entropy

The **entropy** of a probability distribution $p_1, \dots, p_n > 0$ is $H(\mathbf{p}) = -\sum_i p_i \ln p_i$. Maximise it subject to $\sum_i p_i = 1$.

$$\frac{\partial H}{\partial p_i} = -\ln p_i - 1 = \lambda \quad \Longrightarrow \quad p_i = e^{-1-\lambda} \quad \text{for every } i.$$

So all p_i are equal: $p_i = \frac{1}{n}$, and $H = \ln n$.

- It is the global maximum: the Hessian of H is $\operatorname{diag}(-1/p_1, \dots, -1/p_n)$, negative definite, so H is strictly concave on the convex constraint set (Lecture 14).
- The uniform distribution is the "most uncertain" one. Adding the constraint $\sum_i p_i E_i = E$ gives instead $p_i \propto e^{-\beta E_i}$: the Boltzmann distribution of statistical physics.

Eigenvalues as constrained extrema

For a symmetric matrix A: the extrema of $f(\mathbf{x}) = \mathbf{x}^\top A\mathbf{x}$ on the sphere $\mathbf{x}^\top\mathbf{x} = 1$.

$\nabla f = 2A\mathbf{x} = \lambda \cdot 2\mathbf{x} = \lambda\nabla g$ means $A\mathbf{x} = \lambda\mathbf{x}$: the candidates are the unit **eigenvectors**, and there $f = \mathbf{x}^\top(\lambda\mathbf{x}) = \lambda$.

> **Conclusion**
>
> On the unit sphere, the maximum of $\mathbf{x}^\top A\mathbf{x}$ is the largest eigenvalue of A and the minimum the smallest, attained at corresponding eigenvectors.

- The maximum exists (the sphere is closed and bounded): so every symmetric matrix has a real eigenvalue.
- PCA: the direction of largest variance is the top eigenvector of the covariance matrix (Linear Algebra, Unit 23).

Extrema on a closed region

Example ($f = x^2 + 2y^2 - x$ on the disc $x^2 + y^2 \le 1$)

Interior. $\nabla f = (2x - 1,\ 4y) = \mathbf{0}$ at $\left(\frac{1}{2}, 0\right)$, $f = -\frac{1}{4}$.

Boundary. $(2x - 1,\ 4y) = \lambda\,(2x, 2y)$. The second equation: $y = 0$ or $\lambda = 2$.

- $y = 0$: $(\pm 1, 0)$, $f(1, 0) = 0$, $f(-1, 0) = 2$.
- $\lambda = 2$: $2x - 1 = 4x$, so $x = -\frac{1}{2}$, $y = \pm\frac{\sqrt{3}}{2}$, $f = \frac{1}{4} + \frac{3}{2} + \frac{1}{2} = \frac{9}{4}$.

Compare $-\frac{1}{4}$, 0, 2, $\frac{9}{4}$: maximum $\frac{9}{4}$ at $\left(-\frac{1}{2}, \pm\frac{\sqrt{3}}{2}\right)$, minimum $-\frac{1}{4}$ at $\left(\frac{1}{2}, 0\right)$.

Lecture 14's recipe, with Lagrange multipliers for the boundary.

05

More constraints, and limits of the method

Several constraints

Theorem

If $\mathbf{a}$ is a local extremum of f subject to $g_1 = c_1, \dots, g_k = c_k$, and $\nabla g_1(\mathbf{a}), \dots, \nabla g_k(\mathbf{a})$ are linearly independent, then

$$\nabla f(\mathbf{a}) = \lambda_1 \nabla g_1(\mathbf{a}) + \cdots + \lambda_k \nabla g_k(\mathbf{a}).$$

Example (Extrema of z on the ellipse $x^2 + y^2 = 1$, $x + y + z = 1$)

$(0, 0, 1) = \lambda\,(2x, 2y, 0) + \mu\,(1, 1, 1)$. Third component: $\mu = 1$; then $2\lambda x = 2\lambda y = -1$, so $x = y = \pm\frac{1}{\sqrt{2}}$ and $z = 1 \mp \sqrt{2}$.

Maximum $1 + \sqrt{2}$ at $\left(-\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}, 1 + \sqrt{2}\right)$, minimum $1 - \sqrt{2}$ at $\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 1 - \sqrt{2}\right)$.

Inequality constraints (outlook)

Minimise f subject to $g(\mathbf{x}) \le c$. At a minimum $\mathbf{a}$, either

- $g(\mathbf{a}) < c$: the constraint is **inactive**, and $\nabla f(\mathbf{a}) = \mathbf{0}$; or
- $g(\mathbf{a}) = c$: it is **active**, and $\nabla f(\mathbf{a}) = -\mu \nabla g(\mathbf{a})$ with $\mu \ge 0$ — f must not decrease towards the inside, where $g < c$.

Both cases together: $\nabla f + \mu \nabla g = \mathbf{0}$, $\mu \ge 0$, $\mu\,(g - c) = 0$.

> **Remark**
>
> These are the **Karush–Kuhn–Tucker** (KKT) conditions, for one constraint. With many constraints they are the foundation of constrained optimisation algorithms, and of the support vector machine in machine learning.

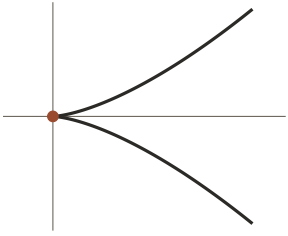
When the method fails

Example (Minimise $f = x$ on the curve $y^2 = x^3$)

On the curve $x = |y|^{2/3} \geq 0$, so the minimum is 0, at $(0, 0)$.

But with $g = y^2 - x^3$: $(1, 0) = \lambda\,(-3x^2,\ 2y)$. The second component gives $\lambda = 0$ (then $1 = 0$) or $y = 0$, so $x = 0$ (then $1 = 0$ again). **No solution.**

At $(0, 0)$, $\nabla g = \mathbf{0}$: the curve has a cusp, and the theorem does not apply. That is why step 2 of the recipe is there.



The cusp $y^2 = x^3$.

06

Summary and practice

Summary

- **Lagrange**: at an extremum of f on $\{g = c\}$ with $\nabla g \neq \mathbf{0}$, $\nabla f = \lambda \nabla g$ — the level set of f is tangent to the constraint.
- Recipe: solve $\nabla f = \lambda \nabla g$, $g = c$; add the points with $\nabla g = \mathbf{0}$; compare values (on a closed bounded set).
- $\lambda = M'(c)$: the rate at which the optimum changes with the constraint level.
- Applications: the square, AM–GM, maximum entropy (uniform), eigenvalues as extrema of $\mathbf{x}^\top A\mathbf{x}$ on the sphere.
- Several constraints: $\nabla f = \sum_j \lambda_j \nabla g_j$. Inequalities: KKT.

Practice

1. Find the extrema of xy on the circle $x^2 + y^2 = 8$.
2. Find the point of the plane $x + 2y + 2z = 9$ closest to the origin.
3. A closed box has surface area 6. Find the largest possible volume.
4. Find the extrema of $2x + y$ on the ellipse $x^2 + 4y^2 = 4$.
5. Find the distribution (p_1, p_2, p_3) of largest entropy, and the entropy.
6. Find the extrema of $\mathbf{x}^\top A\mathbf{x}$ on the unit circle for $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$.
7. Let $M(c)$ be the maximum of xy on $x^2 + y^2 = c$. Find $M(c)$ and λ, and check that $M'(c) = \lambda$.

Answers

1. $y = 2\lambda x$, $x = 2\lambda y$ give $x^2 = y^2 = 4$. Maximum 4 at $\pm(2, 2)$, minimum -4 at $\pm(2, -2)$.
2. $\nabla(x^2 + y^2 + z^2) = \lambda\,(1, 2, 2)$: the point $(1, 2, 2)$, at distance 3.
3. $xy + yz + zx = 3$; the equations force $x = y = z = 1$: volume 1 (a cube).
4. $x = \frac{1}{\lambda}$, $y = \frac{1}{8\lambda}$, $\lambda = \pm\frac{\sqrt{17}}{8}$: maximum $\sqrt{17}$, minimum $-\sqrt{17}$.
5. The uniform distribution $\left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right)$, $H = \ln 3$.
6. Eigenvalues 3 and 1: maximum 3 at $\pm\frac{1}{\sqrt{2}}(1, 1)$, minimum 1 at $\pm\frac{1}{\sqrt{2}}(1, -1)$.
7. At the maximum $x = y$, so $M(c) = \frac{c}{2}$ and $\lambda = \frac{1}{2} = M'(c)$.

Next time

Lecture 17 — Implicit and inverse function theorems (Unit 17)

- When does $g(x, y) = c$ define y as a function of x?
- The implicit function theorem, and implicit differentiation
- The inverse function theorem: local invertibility from the Jacobian

Reading: Marsden & Tromba, *Vector Calculus*, Section 3.5.

Questions: the module channel on Discord, or office hours, Monday–Friday, 10:00–17:00 (Baku time, UTC+4), ali.khudiyev@wosm.academy.